

Research article

The effect of chemical and organic N inputs on N₂O emission from rain-fed crops in Eastern MediterraneanMichalis Omirou^{a,*}, Ioannis Anastopoulos^b, Dionysia A. Fasoula^c, Ioannis M. Ioannides^{a,**}^a Department of Agrobiotechnology, Agricultural Research Institute, P.O.Box 22016, Nicosia, 1516, Cyprus^b University of Cyprus, Department of Chemistry, P.O.Box 20537, Nicosia, 1678, Cyprus^c Department of Plant Breeding, Agricultural Research Institute, P.O.Box 22016, Nicosia, 1516, Cyprus

ARTICLE INFO

Keywords:

Climate change
Nutrient management
N₂O
Emission factors

ABSTRACT

Nitrogen has a significant contribution to global warming and its reduction in agriculture is expected to reduce N₂O emissions having however adverse effects on the productivity of agricultural ecosystems. Maintaining systems productivity with alternative N sources i.e manure and composts could be a strategy also to mitigate N₂O emissions. In this paper, we present the effect of different N sources (organic and chemical) on field N₂O emissions and how these emissions are associated with soil available N forms (NH₄⁺ and NO₃⁻) in three different rain-fed crops namely barley, pea and vetch grown in Cyprus for two growing seasons. The daily emissions ranged from -3.11 to 12.3 g N-N₂O/ha/day, while cumulative emissions ranged from 119 g N-N₂O/ha to 660 g N-N₂O/ha depending on crop and nitrogen source type. The emissions showed a seasonal pattern and WFPS has been identified as a critical soil parameter controlling daily N₂O emissions. The daily N₂O fluxes in the current study derives mainly from nitrification irrespectively crop type or nitrogen source type. Specific emission factors for each crop cultivated under different N source type were calculated and ranged from 0.03% ± 0.02–0.34% ± 0.09. The application of manure and chemical fertilizers cause similar intensity of N₂O emissions while compost exhibited the lower emission factors. These findings suggest that composts could be integrated in a nutrient management strategy of rain-fed crops with less N₂O emissions. The high background emissions found suggest also that other factors than external inputs are associated with N₂O emissions and further studies including the response of microbial community structure and their contribution and association with N₂O emissions.

1. Introduction

Rainfed agricultural systems remain an essential tool for tackling food security since they remain the largest agricultural utilized area in semi-dry agroecological zones. These systems are characterized by low soil fertility, and their productivity relies on the frequency and the amount of the annual rainfall. (Bhattacharya, 2019). Climate change is expected to put additional pressure on these agricultural areas with adverse effects on the production and the quality of the products. Improving the fertility of these soils to maintain or even increase the productivity of the agricultural ecosystem remains a challenge.

Numerous studies showed the positive effect of green and animal manure on plant performance and the physicochemical properties of marginal soils (Diacono et al., 2019; Iqbal et al., 2019; Kranz et al., 2020). The application of organic amendments presents multiple

benefits for agricultural systems since it offers essential plant elements, significantly improves the physico-chemical properties of the soils, and, subsequently, enhances soil fertility through the increase of organic carbon (Bastida et al., 2015; Bronick and Lal, 2005; Diacono and Montemurro, 2010). The use of organic amendments besides their positive effects has also been associated with N₂O emissions (Thangarajan et al., 2013), which are, in some cases, higher than those derived after chemical fertilization. For example, the application of animal manure resulted in significantly higher N₂O emissions compared to the chemical in an irrigated cotton field under extremely arid conditions (Kuang et al., 2018). Likewise, the emissions from maize received slurry were substantially higher from the plots received chemical fertilizers in two different soils under temperate conditions (van Groenigen et al., 2004). In contrast, the application of organic and inorganic fertilizers in rain-fed grown barley, as well as irrigated maize, caused similar

* Corresponding author.

** Corresponding author.

E-mail addresses: michalis.omirou@ari.gov.cy (M. Omirou), ioannide@ari.gov.cy (I.M. Ioannides).

cumulative emissions without any significant differences (Halvorson et al., 2016; Meijide et al., 2009).

These contradictions are associated with the inherent characteristics of organic amendments, the site-specific environmental conditions of the soil, and of course, the crop type. Comparing the amount of N₂O emitted from soils received different types of organic amendments (animal and green manures, composts, biochar) remains challenging and involves significant difficulties in understanding the mechanisms responsible for any differences observed on the N₂O emissions (Aguilera et al., 2013). For example, the higher N₂O emissions measured in soils treated with fresh or composted manure solids than that of biochar has been attributed to the lower available C and N content (Zhu et al., 2014). Also, the application of pig, poultry and cow manure in the soil resulted in different cumulative N₂O emission rates that might be governed by NH₄⁺ concentration, rather than by total N concentration or the dissolved organic carbon (Kim et al., 2019). A lab incubation study showed that plant residues with a low C/N ratio might result in substantially higher N₂O emissions (Li et al., 2016). Recently we also demonstrated that the type of organic amendment is a crucial factor controlling the N₂O emissions from soil (Christodoulou et al., 2019).

Studies regarding the effect of applying different organic amendments on soil N₂O emissions in various rain-fed crops in the Mediterranean like climates are limited (Barton et al., 2016; Heller et al., 2010; Meijide et al., 2009). A better understanding of N₂O fluxes from rain-fed systems in semi-arid regions would be gained by including more measurements and improving the representation of different types of N sources used in different agricultural systems (Stehfest and Bouwman, 2006).

The N₂O emissions from rain-fed agricultural ecosystems in Cyprus have never been studied and there is an urgent need to examine whether the emissions under the pedoclimatic conditions differs across crop types and type of nitrogen inputs. Thereby, the objectives of the current study were to examine the impact of different organic types of N inputs on N₂O emissions, and compare them with the emissions generated after chemical fertilization in three different rain-fed crops in two year field experiment. To better understand the direct N₂O emissions from the soil their association with soil parameters was examined. Finally, we calculated country-specific emission factors for each rain-fed crop and N management scheme for the first time in Cyprus.

2. Material and methods

2.1. Site description and experimental design

The current study was performed in the Experimental Station of Agricultural Research Institute at Acheleia Paphos. The soil using Bouyoukos method consisted of 22% sand, 26% silt, and 52% clay with pH (1:2.5 soil: water slurry) value of 8.4. The total organic C (Walkley-Black method), N (Kjeldahl method), and nitrate-nitrogen (cadmium reduction method) were 17.1 and 0.89 g/kg and 4.78 µg/g, respectively. The CaCO₃ (Bernard Method), electrical conductivity EC (1:2.5 soil: water slurry), and P (Oslen) were estimated to be 1.51%, 0.27 dS/m, and 3.72 mg/kg, respectively. Soil water holding capacity (%) was found to be 56%. Different types of nutrient inputs were examined in three crop species namely, barley, vetch and pea suited for the rain-fed conditions of the Eastern Mediterranean region to monitor direct N₂O emission. In detail, experimental plots received organic and chemical fertilization with the following rates following the guidelines of good agricultural practices: i) goat composted manure (MAN) with C/N equal to 14.6 was applied at the rate of 6 ton/ha (101 kg N/ha) at the end of November each year and ii) compost (COM) derived from plant residues with C/N equal to 24.5 was applied at the rate of 9.6 ton/ha (111 kg N/ha) at the end of November each year. Chemical fertilization was achieved using ammonium nitrate (NH₄NO₃) and the experimental plots received ammonium nitrate (F) at the rate of 180 kg/ha (60 kg N/ha). In vetch-seeded plots, the biomass was incorporated into the soil during

flowering using a rotavator. Before incorporation, the biomass produced in 1 m² was weight and a subsample of 300 g was used to determined N and dry biomass. Plots without the application of any external nitrogen input were used as negative control (C) treatments for all crops under study to determine the emission factors from each crop and each nutrient management scheme. All treatments were arranged in a randomized complete block design with three replicates at 21 m² plots and repeated in the same plots during two consecutive growing seasons. All crops were seeded on 17 December in 2016, and on 21 December in 2017, the barley cultivar “Achna” was sown at the seed rate of 120 kg/ha while a local cultivar of vetch and pea were sown at the seed rate of 150 kg/ha.

2.2. Direct soil N₂O measurements

In situ soil, N₂O fluxes were measured using static chamber method. Flux measurements were taken over the period 01/12/2016 to 15/06/2018 at a weekly or 15 days intervals. The chambers were calibrated during the winter period to determine the limit of flux detection following the methodology described previously (Parkin et al., 2012). In each experimental plot, a static chamber was deployed to reduce the within plot variability. In detail, immediately after sowing, a stainless still cylindrical chamber base covering 0.132 m² was inserted into the soil about 10 cm deep. The stainless-steel chamber had a total volume of 31.7 L was tightly fitted to the top of the base. Samples were taken every 15 min from each chamber for 1 h using a 30 ml syringe connected by a two-way stopcock and immediately transferred in pre-evacuated Exetainer vials (5.7 ml) equipped with butyl rubber stoppers. Gas samples are loaded in an Agilent Chromatograph (Agilent 5975, MSD) linked with a mass spectrometer using a capillary column RQ-Plot (Restek) as previously described (Ekeberg et al., 2004).

2.3. Calculation of daily and cumulative N₂O fluxes

Nitrogen N₂O emissions were calculated using the equation below:

$$N - N_2O = \frac{dC}{dt} \times \frac{V_c}{A_c} \times \frac{28}{M_v} \times 60 \quad (1)$$

where N–N₂O is the emission flux of N₂O in µg N₂O–N m^{−2} h^{−1}, dC/dt is the rate of change in N₂O concentration (ppmv.min^{−1}) in the chamber headspace, V_c the volume of the chamber (L), A the area covered by the chamber (m²), 28 is the molecular mass of N in N₂O (g.mol^{−1}) and M_v is the molecular volume of gas at chamber temperature (L.mol^{−1}). Hourly fluxes then converted to daily fluxes (g N–N₂O/ha/day). Cumulative flux emissions over the experimental period were calculated using trapezoidal integration. The soil direct N₂O emission factors in different crops were calculated as the difference between the N–N₂O emitted from the soil received the different nutrient inputs (manure, compost and ammonium nitrate) from the control plots divided by the amount of N applied within the treatment (Barton et al., 2016).

2.4. Soil analysis

Three soil samples were collected from each plot at 0–15 cm at each N₂O emission measurement and then mixed thoroughly and stored at 4 °C until analysis. Soil nitrates and ammonium were extracted with 1 M KCl using a 1:5 soil:extractant ratio. Nitrates were determined by spectrophotometry after reduction to nitrite using a cadmium column and NH₄⁺ using the indophenol blue method (Christodoulou et al., 2019). Daily average temperature and rainfall were collected from the meteorological weather station of the ARI Experimental Station. The water-filled pore space (WFPS) is calculated according to the following equation currently used to compare the level of anoxia in different soils, which is one of the main factors controlling N₂O emissions. The WFPS is calculated according to Eq

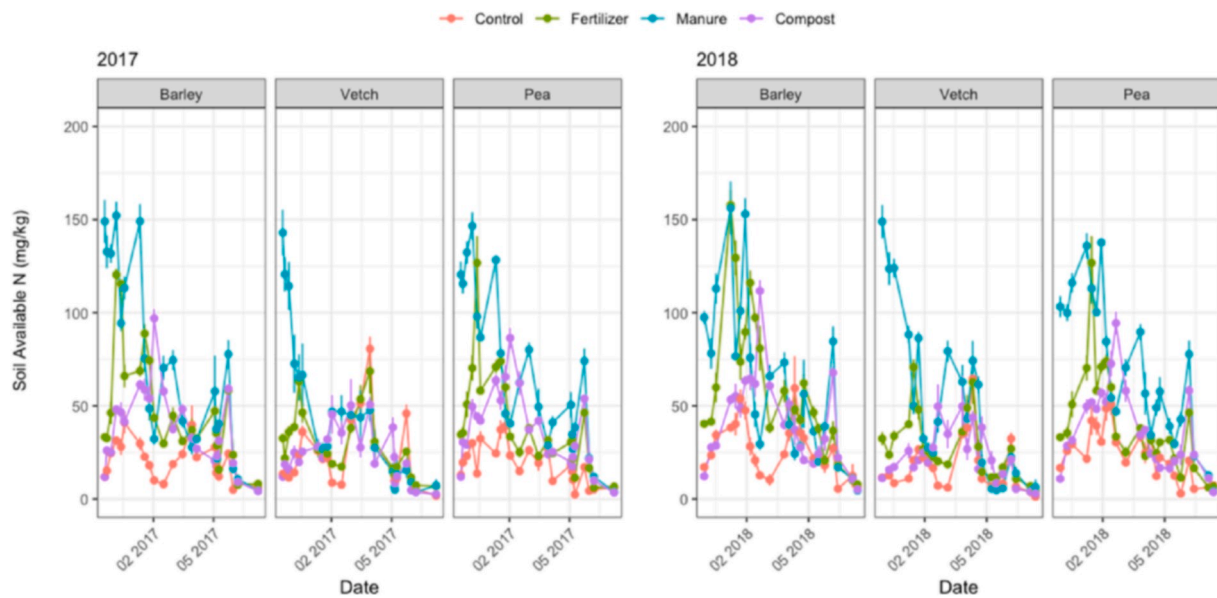


Fig. 1. Evolution of total available nitrogen (mg/kg dry soil), (sum of NO_3^- and NH_4^+) in two growing seasons in different crops barley, pea and vetch grown under different type N sources. Error bars denotes standard error of the mean.

$$\text{WFPS (\%)} = 100 \times \frac{\text{SM}}{1 - (\text{BD}/2.65)} \quad (2)$$

where SM is the volumetric soil moisture (g g^{-1} dry soil), BD the soil bulk density (1.25 g cm^{-3}) and 2.65 is the soil particle density (g cm^{-3}).

2.5. Statistical analysis

General linear models were used to analyze the effects of the treatments and crops on cumulative emissions using the R software base commands (R Core Team, 2019). *Post hoc* comparisons between the different treatments and the controls were performed using Least

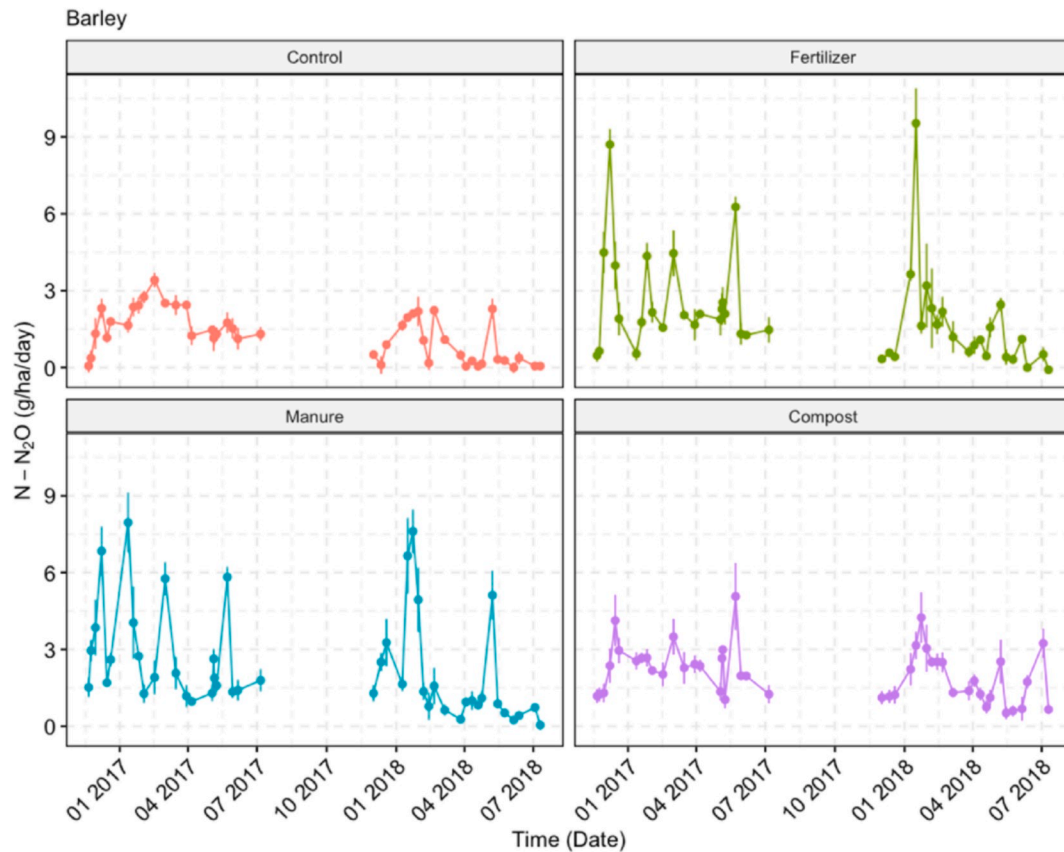


Fig. 2. Daily $\text{N-N}_2\text{O}$ emissions (g/ha/day) throughout two growing seasons (2016–2017 and 2017–2018) for barley grown under different type N sources. Error bars denotes standard error of the mean.

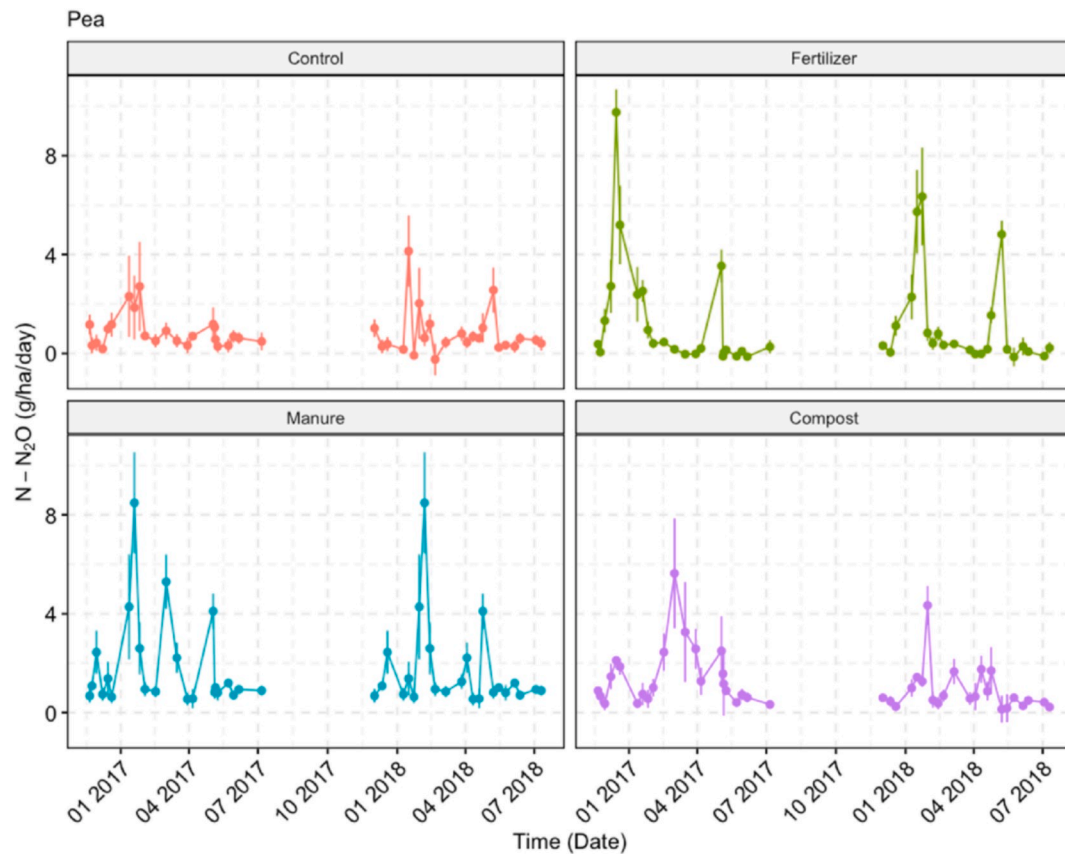


Fig. 3. Daily N–N₂O emissions (g/ha/day) throughout two growing seasons (2016–2017 and 2017–2018) for pea grown under different type N sources. Error bars denotes standard error of the mean.

Significance difference test at $\alpha = 0.05$. The effect of the different treatments in each crop on daily nitrous oxide fluxes, as well as N available soil forms, was assessed using mixed effect models and lmer4 package (Bates et al., 2010). Type of nutrient management scheme and time were used as fixed while experimental plots as random factors. When necessary, N₂O fluxes were log-transformed to achieve a normal distribution that was checked using the Shapiro-Wilk test. The relationship between soil available nitrogen forms, WFPS, and N–N₂O emissions during the time was assessed using the Pearson correlation test.

3. Results

3.1. Changes in nitrogen available forms in different crops and nutrient management schemes over growing seasons

The time during the growing season, crop type, N source type as well as their interactions, had a significant impact on soil available nitrogen forms ($F = 6.95$, $p < 0.01$). Soils received no inputs (Control) in all crops exhibited low levels of N–NO₃ and N–NH₄ throughout the growing season except plots cultivated with vetch (data not shown). In these plots, a gradual increase of available soil N was observed from late February to late March (Fig. 1) in both years, which coincided with the soil incorporation of the above-ground biomass of the crop. The application of organic amendments (manure or compost) as well as ammonium nitrate caused a substantial increase of soil available nitrogen in all crops both in 2017 and 2018 compared to control plots (Fig. 1). Only at the end of the growing season, the concentration of available soil nitrogen was similar between plots received external nutrient input, and those received no input in all crops.

3.2. Nitrous oxide fluxes in different crops grown under different nutrient management schemes

Gas samples were taken at regular intervals from 21/11/2016 to 07/07/2017 for the growing season 2016–2017 and from 02/12/2017 to 10/07/2018 for the growing season 2017–2018 in all crops grown under different nutrient management schemes. Barley, pea and vetch exhibited different patterns of daily soil nitrous oxide fluxes throughout the growing seasons (Figs. 2–4). The N₂O flux pattern within each crop was depended on the type of nutrient management followed. In all crops, the lowest flux range was observed in plots received no inputs, followed by plots received compost (Figs. 2–4). In detail, daily fluxes in barley received no input ranged from -0.6 to 3.9 g/ha/day depending on the time within each growing season. The mean daily emission measured in barley without any nutrient inputs was 1.2 g/ha/day. Both in 2017 and 2018, in barley plots received external inputs, N₂O emissions peaks have been associated positively with soil water content (Fig. 2), and their intensity was depended on the type of input. For example, the emissions noticed in plots received manure, and ammonium nitrate was substantially higher compared to those noticed in plots received compost (Fig. 2). A similar trend was also noticed in pea and vetch, where N₂O emissions have been driven by N addition during both growing seasons (Figs. 3 and 4). In pea, the mean daily N₂O emission was 0.8 g/ha/day in control plots, 1.2 g/ha/day in plots received compost and ammonium nitrate, and 1.7 g/ha/day in manure treated plots. Vetch cultivated plots exhibited higher mean daily emissions ranged from 1.2 to 2.3 g/ha/day. The highest N₂O daily emissions have been noticed after the incorporation of the above-ground biomass into the soil, and this was during February and March (Fig. 4).

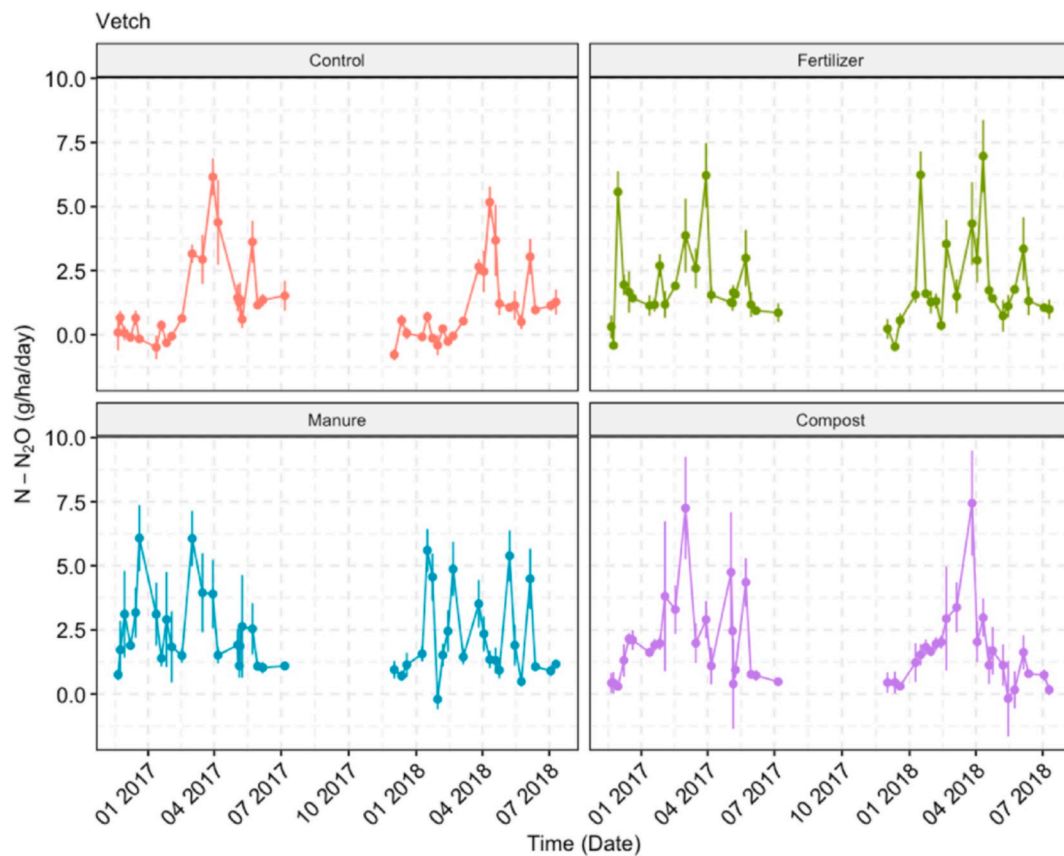


Fig. 4. Daily N-N₂O emissions (g/ha/day) throughout two growing seasons (2016–2017 and 2017–2018) for vetch grown under different type N sources. Error bars denotes standard error of the mean.

3.3. Cumulative N₂O emissions in different crops and nutrient management schemes

Both during 2017 and 2018, cumulative N-N₂O emissions were significantly affected by type of N source, crop and their interactions ($F = 3.09$, $p < 0.001$). Overall, cumulative emissions were lower in 2018 than in 2017 in all crops and N source type. In detail, in barley, N₂O emissions from control soil was 417 g N₂O-N ha⁻¹, and was significantly lower than the 564, 534 and 649 g N₂O-N ha⁻¹ recorded in fertilizer, compost and manure treated soils during 2017 respectively. On the contrary, during 2018, N₂O emissions from control soils, was 180 g N₂O-N ha⁻¹ while in soils treated with fertilizer, compost, and manure, the cumulative emissions were 326, 393 and 395 g N₂O-N ha⁻¹ respectively (Fig. 5). During 2017, in pea, the cumulative emissions were always lower than plots cultivated with barley and vetch irrespectively the type of N implemented (Fig. 5). In contrast, in 2018 the emissions from pea plots were different from those measured in barley and vetch depending on the nutrient type applied (Fig. 5). For example, cumulative emissions in pea-cultivated plots were lower than those measured in barley plots only when compost was used (Fig. 5). In plots received fertilizer and manure, the cumulative emissions were significantly higher in vetch than pea, while no differences were noticed between barley and vetch plots (Fig. 5).

3.4. Emission factors and nutrient management schemes

The percentage of nitrogen lost as N₂O emissions relative to the different types of nitrogen inputs are presented in Table 1. Overall, the amount of N that is emitted from all treatments as N₂O is very low, as estimated from the emission factors (EF). The mean of emission factors calculated under the rain-fed agricultural crops evaluated in the current

study irrespectively the type nutrient source was 0.19% in two consecutive growing seasons. The lowest emissions factors have been noticed in compost treated plots compared to soils received fertiliser or manure. This is the first study performed in different rain-fed crops in Cyprus, and it is evident that direct soil emissions of N₂O are mainly influenced by the environmental conditions as well as the type of N introduced into the system.

4. Discussion

4.1. The effect of different N types on N₂O emissions

In the current field study, we tested the effect of different N inputs on barley, vetch and pea under the semi-arid climatic conditions of Cyprus. The mean daily N-N₂O emissions measured from the different rain-fed crops ranged from -3.11 to 12.3 g N-N₂O/ha/day and are similar to those reported in other agricultural ecosystems located in Mediterranean like environment (Aguilera et al., 2013; Barton et al., 2011; Gregorutti and Cavaglia, 2017). Peaks of N-N₂O emissions have been observed during winter and after the application of external inputs (Figs. 2–4). It has been previously published that the N₂O pulses are stimulated after the application of N external inputs into the soil and this is related to nitrogen and water availability (Forte et al., 2017; Millar et al., 2018). Indeed, N availability was significantly higher in manure, and ammonium nitrate treated soils in all crop's contrary to compost and control soils, which exhibited significantly lower content of N available forms (Fig. 1).

The negative and zero N₂O fluxes have been measured, further suggests that soil could act as an N₂O sink. The negative N₂O fluxes observed occasionally in all treatments could be the result of high soil water content, which in turn triggers the production of N₂ (Flechar

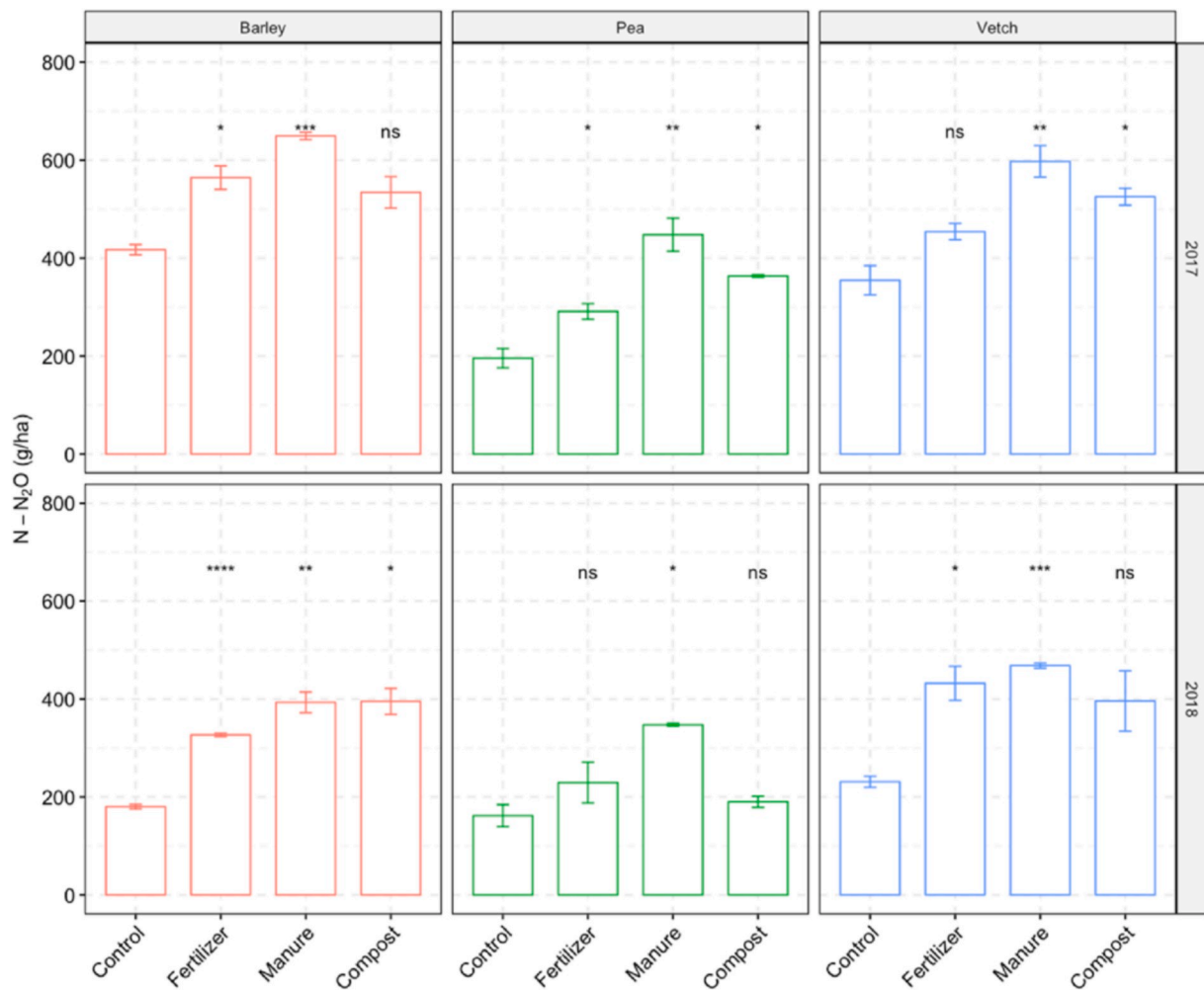


Fig. 5. Seasonal cumulative N-N₂O emissions (g/ha) in barley, pea and vetch grown under different N type sources. Error bars denote standard error of the mean and asterisk (* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$) indicate statistically significant differences between cumulative emissions derived from the different N treatments with the background emissions (Control), ns indicates not significant differences.

Table 1

Emissions factors calculated for barley, pea and vetch plots during two growing seasons (2016–2017 and 2017–2018) grown under different N type sources, namely ammonium nitrate (Fertilizer), goat composted manure (Manure) and composted plant residues (Compost).

Growing season	Crop	Fertilizer	Manure	Compost
2016–17	Barley	0.25 ± 0.09 ¹	0.23 ± 0.026	0.11 ± 0.049
	Pea	0.16 ± 0.11	0.25 ± 0.032	0.15 ± 0.034
	Vetch	0.17 ± 0.065	0.24 ± 0.105	0.15 ± 0.037
2017–18	Barley	0.24 ± 0.023	0.21 ± 0.030	0.19 ± 0.042
	Pea	0.11 ± 0.06	0.18 ± 0.037	0.03 ± 0.021
	Vetch	0.34 ± 0.096	0.23 ± 0.027	0.15 ± 0.079
Overall mean		0.21 ± 0.0832	0.22 ± 0.05	0.13 ± 0.068
	Barley mean	0.245 ± 0.0593	0.221 ± 0.027	0.15 ± 0.063
	Pea mean	0.135 ± 0.079	0.215 ± 0.047	0.09 ± 0.07
	Vetch mean	0.255 ± 0.118	0.235 ± 0.069	0.15 ± 0.055

Standard Error of the mean emission factor ¹ $n = 3$, ² $n = 18$, ³ $n = 6$.

et al., 2005), and this could be the case during the winter period in the current study. Also, the highest negative fluxes have been noticed in control and compost treated soils and these could be linked with the low concentration of available nitrogen into the soil. In compost treated soil, the negative fluxes noticed could also be associated with the addition of carbon into the soil, which stimulates microbial respiration leading to anaerobic microsites where N₂ is produced. It has been proposed that aerobic denitrification could be another possible route of N₂O consumption in soils leading to negative fluxes (Chapuis-Lardy et al., 2007). Possibly, the low water content observed, combined with the low N availability and the increase of organic C after the addition of the compost could shift the structure and the abundance of N₂O-reducing microbial community. It has been reported that the relative abundance of N₂O-reducing bacteria varied among soil and environmental conditions (Jones et al., 2013). Recently, we perceived that organic amendments altered the abundance of *nosZII* bacterial community (Anastopoulos et al., 2019) while others showed that the incorporation of biochar increased the microbial N₂O reduction activity, thereby leading to lower N₂O emissions (Harter et al., 2017). Besides any changes in microbial community structure and activity, the substantially lower emissions flux observed in compost treated compared to the manure treated soil could partially related to the different characteristics of these materials. The C/N ratio of the compost was higher (data not shown) compared to that of manure suggesting lower N availability. Previous studies suggested that an increase in the C/N ratio of organic

Table 2

Pearson correlations and statistical significance between daily N₂O emissions and measured soil parameters. Asterisks (* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$) indicate statistically significant correlations.

	N source type	WFPS (%)	NO ₃ ⁻ (mg/kg)	NH ₄ ⁺ (mg/kg)
Barley, 2017				
N-N ₂ O (g/ha/day)	Fertilizer	0.40***	0.42***	0.22**
	Manure	0.38***	0.41***	0.15**
	Compost	0.39***	0.42***	0.21***
	Control	0.37***	0.45***	0.21**
Barley, 2018				
N-N ₂ O (g/ha/day)	Fertilizer	0.40***	0.41***	0.22**
	Manure	0.41***	0.40***	0.16*
	Compost	0.39***	0.38***	0.19**
	Control	0.35***	0.42***	0.23**
Pea, 2017				
N-N ₂ O (g/ha/day)	Fertilizer	0.34***	0.40***	0.21**
	Manure	0.37***	0.40***	0.16*
	Compost	0.33***	0.42***	0.18**
	Control	0.33**	0.44***	0.19**
Pea, 2018				
N-N ₂ O (g/ha/day)	Fertilizer	0.35***	0.36***	0.20**
	Manure	0.37***	0.37***	0.21**
	Compost	0.32***	0.32***	0.25***
	Control	0.29***	0.33***	0.21**
Vetch, 2017				
N-N ₂ O (g/ha/day)	Fertilizer	0.31***	0.42***	0.20***
	Manure	0.29***	0.43***	0.15***
	Compost	0.29**	0.39***	0.17**
	Control	0.30***	0.47***	0.17**
Vetch, 2018				
N-N ₂ O (g/ha/day)	Fertilizer	0.30***	0.37***	0.22***
	Manure	0.36***	0.39***	0.11*
	Compost	0.21**	0.31***	0.17***
	Control	0.30***	0.24**	0.27***

amendments could lead to a significant decrease in soil direct N₂O emissions (Huang et al., 2004).

Overall, in both growing seasons we found a positive significant correlation between N₂O daily fluxes, NO₃⁻, NH₄⁺, as well as WFPS (Table 2). The N₂O production below 40% WFPS was negligible and substantially increased beyond the threshold of 40% WFPS in all crops. The significant correlation found between the available soil N and N₂O emissions (Table 2) when soil moisture was less than 70% suggests that nitrification is the main route of N₂O emissions (Fig. 6). Most of the N₂O emissions occurred in the range of 40–60% WFPS with maximum emission at approximately 65% WFPS. The optimum WFPS during which max N₂O occurs depends on biological and physical processes that are taking place into the soil during the flux measurements. For example, the nitrification process occurs up to 60% of the WFPS while in higher values, denitrification is the dominant N₂O biosynthetic pathway (Davidson and Verchot, 2000). These findings are further supporting previous reports suggesting that in semi-arid climates, nitrification activity seems to be the main source of N₂O emissions (Barton et al., 2013; Sanz-Cobena et al., 2016).

4.2. N₂O cumulative emissions and emission factors

The background emissions measured during the two growing seasons in all crops and grown under various nutrient management schemes ranged from 49 to 85% of the total emissions measured and ranged from 162 to 417 g N-N₂O/ha. This variability suggests that a large part of the direct soil N₂O emissions cannot be attributed to the application of external N inputs (chemical or organic) but to other factors that are

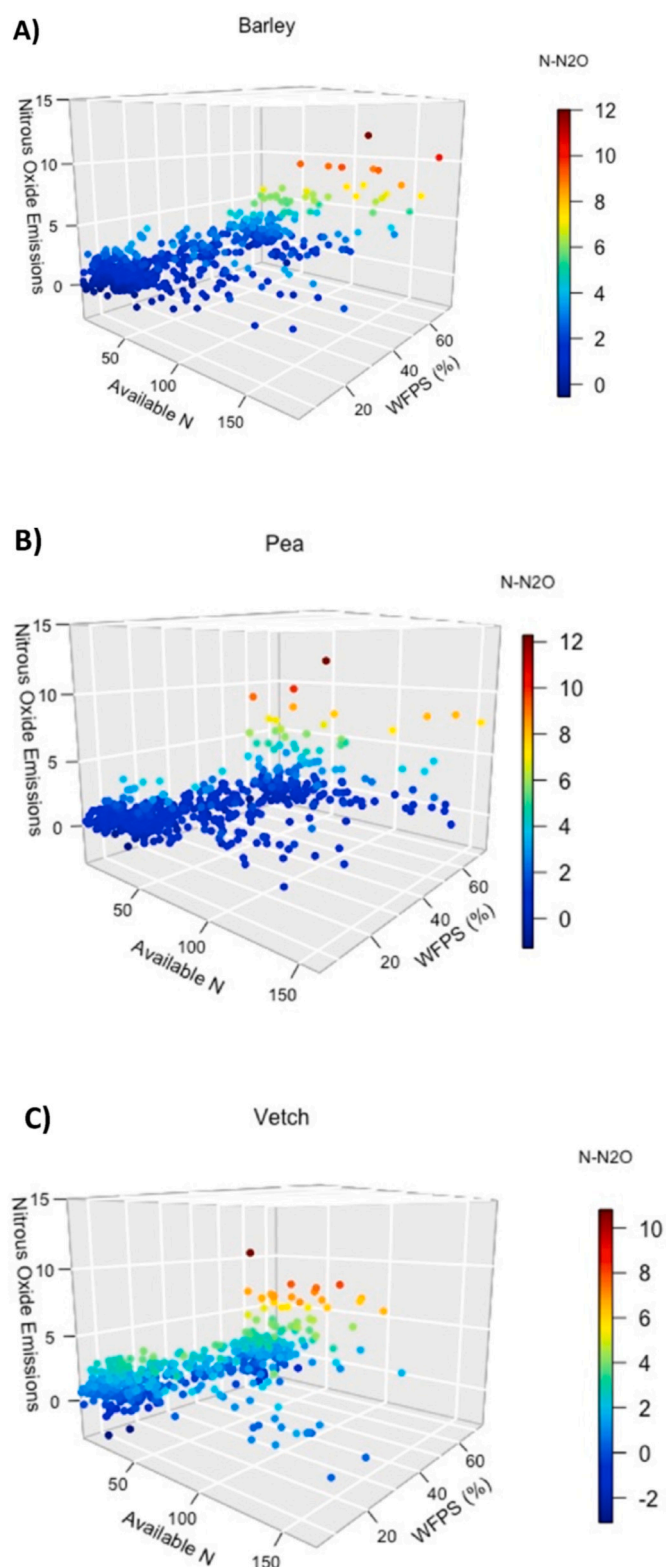


Fig. 6. Daily N-N₂O emissions (g/ha), WFPS (%) and total available nitrogen (mg/kg dry soil), (sum of NO₃⁻ and NH₄⁺) measured in A) barley, B) pea and C) vetch in the two growing seasons plotted in the 3D space.

substantially affecting the pathways of N₂O emissions from the soil (Aguilera et al., 2013). Indeed, a strong seasonal effect was noticed, on cumulative emissions between the growing seasons and this was depended on the crop and the type of N inputs ($F = 3.09$, $p < 0.001$). Overall, the emissions during the first growing season were higher

compared to the second growing season. The direct soil N₂O emissions were different between crops and this response depended on the type of nutrient applied. For example, the cumulative emissions derived from pea plots received compost and manure was lower compared to barley and vetch in both growing seasons. In contrast, the application of ammonium nitrate caused higher emissions in barley and vetch during the first growing season and only in vetch plots during the second growing season. Our findings are in line with previous studies showed that under rain-fed conditions the N₂O emissions from grain legumes grown in semi-arid environments are low and their cultivation didn't enhance or further stimulated the emissions (Barton et al., 2011, 2013).

Crop-specific management practices could have a decisive effect on the cumulative N₂O emissions, and this is evident in vetch. Particularly, in vetch plots received no external N inputs, the incorporation of the above-ground biomass caused substantially higher emissions compared to pea (in both growing season) and barley during the second growing season only. This is line with previous studies showed that the N₂O emissions from legume based green manures could be stimulated (Sanz-Cobena et al., 2014). The majority of these emissions were noticed within the first days after the incorporation of the above-ground biomass and are possibly associated to the increased amount of carbon and nitrogen due to biomass incorporation into the soil (Kandel et al., 2019; Sanz-Cobena et al., 2014). Indeed, after the incorporation of the vetch biomass a significant increase of plant available N was observed (Fig. 1) in both growing seasons. Also, the incorporation of fresh plant biomass stimulates and enhances soil microbial growth and activity leading to a substantial reduction of O₂ levels on soil microsites, thereby driving to higher N₂O emissions (Thangarajan et al., 2013).

The calculated emission factors from all crops nutrient grown under different types of N source ranged from 0.03% ± 0.02–0.34% ± 0.09. These findings are in line with previous studies which demonstrated that the emission factors from agricultural systems in Mediterranean like climates are low (Barton et al., 2016). In the current study, we showed that the calculated EF for each rain-fed crop depends on the type of nutrient incorporated into the soil leading to a significant interaction between crop and nutrient input type ($F = 4.65$, $p < 0.001$). Cayuela et al. (2017) in a meta-analysis review showed that Mediterranean countries should consider using lower N₂O EFs for their inventories due to the idiosyncratic properties and characteristics of agricultural ecosystems in Mediterranean region.

5. Conclusions

The current study reports for the first time cumulative and daily N₂O emissions from three different rain-fed that are important for Mediterranean region in Cyprus. The daily emissions ranged from −3.11 to 12.3 g N-N₂O/ha/day, while cumulative emissions ranged from 119 g N-N₂O/ha to 660 g N-N₂O/ha. The emissions showed a seasonal pattern and WFPS has been identified a critical soil parameter controlling daily N₂O emissions. The source of the daily N₂O fluxes in the current study could be attributed to nitrification irrespectively crop type or nutrient management practice since a significant positive correlation have been observed between N₂O with NO₃[−] and NH₄⁺ and the WFPS was always below 70%. Specific emission factors for each crop cultivated under different nutrient management strategies were calculated. The findings further support previous results suggested that, the default IPCC Tier I emission factor is not suitable for soils in Mediterranean like environments and site-specific emission factors calculations are encouraged. The results of this study demonstrate that under the pedoclimatic conditions of Cyprus, the application of manure and chemical fertilizers cause similar intensity of N₂O emissions. This is particularly useful for the agricultural ecosystems of the country as well as the region of Eastern Mediterranean since the application of manure has multiple benefits in the soil as well as the productivity of the system in the long term (Diacono and Montemurro, 2010). The application of manure following good agricultural practices (timing, quantity and

incorporation) did not accumulate residual nitrogen in the soil since after the end of the growing season since the level of soil available nitrogen was low (Fig. 1). As a result, the amount of nitrogen forms prone to N₂O were negligible. Interestingly, the N₂O emission factors calculated from plots received compost was substantially lower compared to those measured in plots received chemical and fresh manure N inputs (Table 1). It could be therefore suggested that the use of compost as part of the overall nutrient management strategy of rain-fed crops could eventually lead to lower N₂O emissions. Finally, the high background emissions found suggest also that other factors than external inputs are associated with N₂O emissions and further studies including the response of microbial community structure and their contribution and association with N₂O emissions are needed.

Funds

This work was financially supported by European Union within the LIFE + ORGANIKO (LIFE 14/CCM/CY/000990) project.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

CRediT authorship contribution statement

Michalis Omirou: Writing - original draft. **Ioannis Anastopoulos:** Writing - review & editing.

Acknowledgements

We acknowledge the technical assistance of Mrs Evdokia Neophytou, Mrs Louiza Konstantinou, Mr Feidias Nicolaou and Mr Zacharias Sophocleous.

References

- Aguilera, E., Lassaletta, L., Sanz-Cobena, A., Garnier, J., Vallejo, A., 2013. The potential of organic fertilizers and water management to reduce N₂O emissions in Mediterranean climate cropping systems. A review. *Agriculture, Ecosystems & Environment* 164, 32–52.
- Anastopoulos, I., Omirou, M., Stephanou, C., Oulas, A., Vasiliades, M.A., Efstathiou, A. M., Ioannides, I.M., 2019. Valorization of agricultural wastes could improve soil fertility and mitigate soil direct N₂O emissions. *J. Environ. Manag.* 250, 109389.
- Barton, L., Butterbach-Bahl, K., Kiese, R., Murphy, D.V., 2011. Nitrous oxide fluxes from a grain-legume crop (narrow-leaved lupin) grown in a semiarid climate. *Global Change Biol.* 17, 1153–1166.
- Barton, L., Hoyle, F.C., Stefanova, K.T., Murphy, D.V., 2016. Incorporating organic matter alters soil greenhouse gas emissions and increases grain yield in a semi-arid climate. *Agric. Ecosyst. Environ.* 231, 320–330.
- Barton, L., Murphy, D.V., Butterbach-Bahl, K., 2013. Influence of crop rotation and liming on greenhouse gas emissions from a semi-arid soil. *Agric. Ecosyst. Environ.* 167, 23–32.
- Bastida, F., Selevsek, N., Torres, I.F., Hernández, T., García, C., 2015. Soil restoration with organic amendments: linking cellular functionality and ecosystem processes. *Sci. Rep.* 5, 15550.
- Bates, D., Maechler, M., Bolker, B., 2010. lme4: linear mixed-effects models using Eigen and R. *package version 0.999375-33*. <http://CRAN.R-project.org/package=lme4>.
- Bhattacharya, A., 2019. Chapter 3 - water-use efficiency under changing climatic conditions. In: Bhattacharya, A. (Ed.), *Changing Climate and Resource Use Efficiency in Plants*. Academic Press, pp. 111–180.
- Bronick, C.J., Lal, R., 2005. Soil structure and management: a review. *Geoderma* 124, 3–22.
- Cayuela, M.L., Aguilera, E., Sanz-Cobena, A., Adams, D.C., Abalos, D., Barton, L., Ryals, R., Silver, W.L., Alfaro, M.A., Pappa, V.A., Smith, P., Garnier, J., Billen, G., Bouwman, L., Bondeau, A., Lassaletta, L., 2017. Direct nitrous oxide emissions in Mediterranean climate cropping systems: emission factors based on a meta-analysis of available measurement data. *Agric. Ecosyst. Environ.* 238, 25–35.
- Chapuis-Lardy, L., Wrage, N., Metay, A., Chotte, J.-L., Bernoux, M., 2007. Soils, a sink for N₂O? A review. *Global Change Biol.* 13, 1–17.
- Christodoulou, E., Agapiou, A., Anastopoulos, I., Omirou, M., Ioannides, I.M., 2019. The effects of different soil nutrient management schemes in nitrogen cycling. *J. Environ. Manag.* 243, 168–176.

- Davidson, E.A., Verchot, L.V., 2000. Testing the Hole-in-the-Pipe Model of nitric and nitrous oxide emissions from soils using the TRAGNET Database. *Global Biogeochem. Cycles* 14, 1035–1043.
- Diacono, M., Baldovino-Freitas, P., Serra, F.X.S., 2019. Nitrogen utilization in a cereal-legume rotation managed with sustainable agricultural practices. *Agronomy-Basel* 9, 13.
- Diacono, M., Montemurro, F., 2010. Long-term effects of organic amendments on soil fertility. In: A review, vol. 30. *Agronomy for Sustainable Development*, pp. 401–422.
- Ekeberg, D., Ogner, G., Fongen, M., Joner, E.J., Wickström, T., 2004. Determination of CH₄, CO₂ and N₂O in air samples and soil atmosphere by gas chromatography mass spectrometry, GC-MS. *J. Environ. Monit.* 6, 621–623.
- Flechar, C.R., Neftel, A., Jocher, M., Ammann, C., Fuhrer, J., 2005. Bi-directional soil/atmosphere N₂O exchange over two mown grassland systems with contrasting management practices. *Global Change Biol.* 11, 2114–2127.
- Forté, A., Fagnano, M., Fierro, A., 2017. Potential role of compost and green manure amendment to mitigate soil GHGs emissions in Mediterranean drip irrigated maize production systems. *J. Environ. Manag.* 192, 68–78.
- Gregorutti, V.C., Caviglia, O.P., 2017. Nitrous oxide emission after the addition of organic residues on soil surface. *Agric. Ecosyst. Environ.* 246, 234–242.
- Halvorson, A.D., Del Grosso, S.J., Stewart, C.E., 2016. Manure and inorganic nitrogen affect trace gas emissions under semi-arid irrigated corn. *J. Environ. Qual.* 45, 906–914.
- Harter, J., El-Hadidi, M., Huson, D.H., Kappler, A., Behrens, S., 2017. Soil biochar amendment affects the diversity of nosZ transcripts: implications for N₂O formation. *Sci. Rep.* 7, 3338.
- Heller, H., Bar-Tal, A., Tamir, G., Bloom, P., Venterea, R.T., Chen, D., Zhang, Y., Clapp, C. E., Fine, P., 2010. Effects of manure and cultivation on carbon dioxide and nitrous oxide emissions from a corn field under mediterranean conditions. *J. Environ. Qual.* 39, 437–448.
- Huang, Y., Zou, J., Zheng, X., Wang, Y., Xu, X., 2004. Nitrous oxide emissions as influenced by amendment of plant residues with different C:N ratios. *Soil Biol. Biochem.* 36, 973–981.
- Iqbal, S., Arif, M.S., Khan, H.Z., Yasmeen, T., Thierfelder, C., Li, T., Khan, S., Nadir, S., Xu, J., 2019. Compost amended with N enhances maize productivity and soil properties in semi-arid agriculture. *Agron. J.* 111, 2536–2544.
- Jones, C.M., Graf, D.R.H., Bru, D., Philippot, L., Hallin, S., 2013. The unaccounted yet abundant nitrous oxide-reducing microbial community: a potential nitrous oxide sink. *ISME J.* 7, 417–426.
- Kandel, T.P., Gowda, P.H., Northup, B.K., Rocateli, A.C., 2019. Incorporation and harvest management of hairy vetch-based green manure influence nitrous oxide emissions. *Renew. Agric. Food Syst.* 1–10.
- Kim, S.U., Ruangcharu, C., Kumar, S., Lee, H.H., Park, H.J., Jung, E.S., Hong, C.O., 2019. Nitrous oxide emission from upland soil amended with different animal manures. *Applied Biological Chemistry* 62, 8.
- Kranz, C.N., McLaughlin, R.A., Johnson, A., Miller, G., Heitman, J.L., 2020. The effects of compost incorporation on soil physical properties in urban soils - a concise review. *J. Environ. Manag.* 261, 10.
- Kuang, W., Gao, X., Gui, D., Tenuta, M., Flaten, D.N., Yin, M., Zeng, F., 2018. Effects of fertilizer and irrigation management on nitrous oxide emission from cotton fields in an extremely arid region of northwestern China. *Field Crop. Res.* 229, 17–26.
- Li, X., Sørensen, P., Olesen, J.E., Petersen, S.O., 2016. Evidence for denitrification as main source of N₂O emission from residue-amended soil. *Soil Biol. Biochem.* 92, 153–160.
- Meijide, A., García-Torres, L., Arce, A., Vallejo, A., 2009. Nitrogen oxide emissions affected by organic fertilization in a non-irrigated Mediterranean barley field. *Agric. Ecosyst. Environ.* 132, 106–115.
- Millar, N., Urrea, A., Kahmark, K., Shcherbak, I., Robertson, G.P., Ortiz-Monasterio, I., 2018. Nitrous oxide (N₂O) flux responds exponentially to nitrogen fertilizer in irrigated wheat in the Yaqui Valley, Mexico. *Agric. Ecosyst. Environ.* 261, 125–132.
- Parkin, T.B., Venterea, R.T., Hargreaves, S.K., 2012. Calculating the detection limits of chamber-based soil greenhouse gas flux measurements. *J. Environ. Qual.* 41, 705–715.
- R Core Team, 2019. R: A Language and Environment for Statistical Computing. R Foundation for Statistical Computing, Vienna, Austria.
- Sanz-Cobena, A., Abalos, D., Meijide, A., Sanchez-Martin, L., Vallejo, A., 2016. Soil moisture determines the effectiveness of two urease inhibitors to decrease N₂O emission. *Mitig. Adapt. Strategies Glob. Change* 21, 1131–1144.
- Sanz-Cobena, A., García-Marco, S., Quemada, M., Gabriel, J.L., Almendros, P., Vallejo, A., 2014. Do cover crops enhance N₂O, CO₂ or CH₄ emissions from soil in Mediterranean arable systems? *Sci. Total Environ.* 466–467, 164–174.
- Stehfest, E., Bouwman, L., 2006. N₂O and NO emission from agricultural fields and soils under natural vegetation: summarizing available measurement data and modeling of global annual emissions. *Nutrient Cycl. Agroecosyst.* 74, 207–228.
- Thangarajan, R., Bolan, N.S., Tian, G., Naidu, R., Kunhikrishnan, A., 2013. Role of organic amendment application on greenhouse gas emission from soil. *Sci. Total Environ.* 465, 72–96.
- van Groenigen, J.W., Kasper, G.J., Velthof, G.L., van den Pol-van Dassel, A., Kuikman, P.J., 2004. Nitrous oxide emissions from silage maize fields under different mineral nitrogen fertilizer and slurry applications. *Plant Soil* 263, 101–111.
- Zhu, K., Christel, W., Bruun, S., Jensen, L.S., 2014. The different effects of applying fresh, composted or charred manure on soil N₂O emissions. *Soil Biol. Biochem.* 74, 61–69.